How the plant twin works today

Is the twin physics, data, or both? How were its numbers fitted? Written for readers without a modelling background. Source: 05_twin_2d/app.js (v34), data/sensors.js, data/bends.js and the fit scripts behind them.

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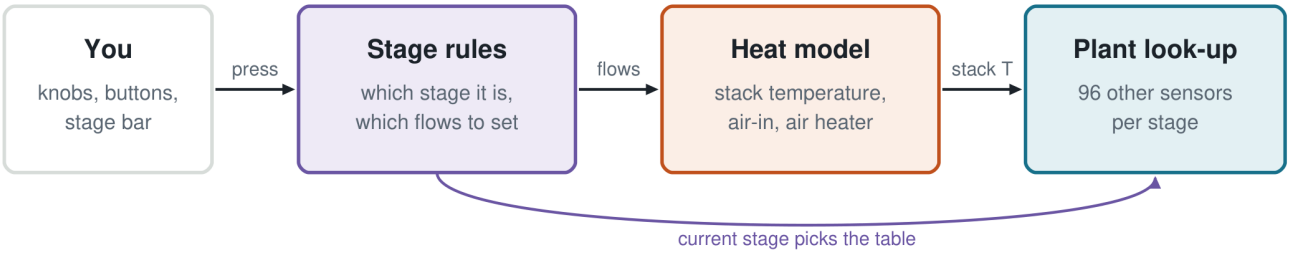
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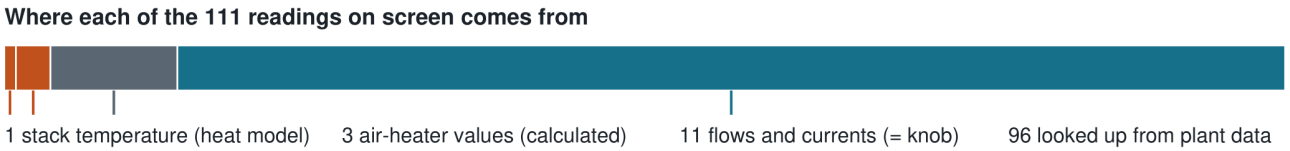
How the plant twin works today

The twin uses **real physics for one thing only**: how hot the stack gets. That physics is a simple heat balance, and its six numbers were **measured from the plant data**, not taken from a textbook. Almost everything else on the screen is **looked up** from what the plant typically showed in the same stage at the same stack temperature.

This matters because it tells you what the twin can be trusted for. It can show how fast the stack warms or cools when you change air, heaters, gas or current. It cannot predict a reading the plant has never shown, because 96 of its 111 readings are repeats of plant history.

Three layers





Your button presses go through stage rules, which set the flows. The heat model turns those flows into a stack temperature. The stack temperature and the stage then pick each other sensor's value from plant look-up tables.

Layer	What it does	Physics?
Stage rules	Reads the stage from the knobs, runs the playbooks (start, load, shut down, trip), locks moves the plant never makes	No: plant practice and your decisions
Heat model	Calculates stack temperature every simulated minute; also air-in temperature and air-heater power	Yes: a heat balance with numbers fitted on plant data
Plant look-up	Gives the other 96 sensors a value for the current stage and stack temperature	No: plant statistics

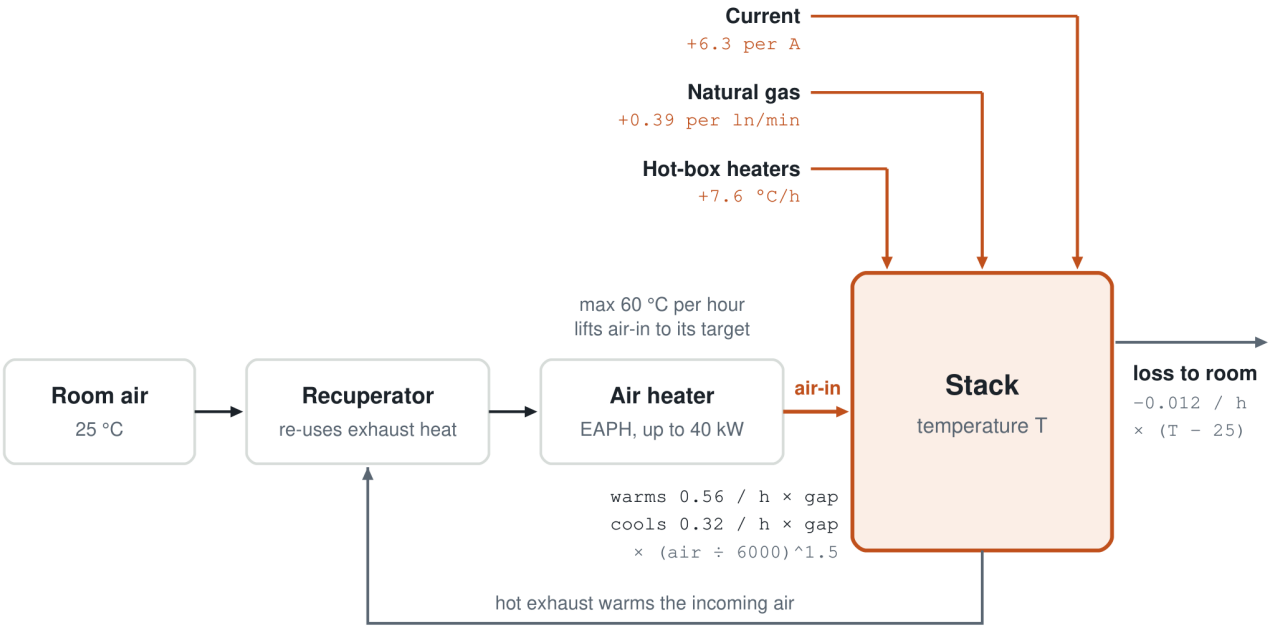
The next pages go through each layer: the physics, how its numbers were fitted, the look-up tables, the stage rules, and how close it gets.

IF YOU REMEMBER ONE THING

The twin calculates the stack's heat (and the air heater). The other 96 readings are the plant's typical values for that stage and temperature.

The physics inside: one heat balance

Think of the stack as a tank of heat. Every simulated minute the twin adds up the heat coming in and the heat going out, and the difference moves the stack temperature up or down.



Room air is warmed twice before it reaches the stack: first by the recuperator, which re-uses the hot exhaust, then by the air heater. Heaters, natural gas and current add heat. The air takes heat away when it is cooler than the stack, and some heat leaks to the room.

The formula, in words

Warming speed of the stack (°C per hour) =
`air term × (air-in temperature − stack temperature) + heaters + gas + current − loss to room`

Term	Number used	Meaning
Air, when warmer than the stack	0.561 per hour × gap	At 6000 ln/min the air closes 56 % of the gap per hour
Air, when cooler than the stack	0.316 per hour × gap	Cools more slowly: the insulated hot box gives back stored heat
More air	× (air ÷ 6000)^1.5	9000 ln/min moves 1.8 times as much heat as 6000
Hot-box heaters on	+7.62 °C per hour	Fixed extra heat while the heaters run
Natural gas	+0.386 °C per hour per ln/min	80 ln/min in fuel switch adds about +31 °C per hour
Current	+6.34 °C per hour per A per string	6.4 A in part load adds about +41 °C per hour
Loss to room	0.0121 per hour × (stack − 25 °C)	At 500 °C about −5.8 °C per hour

Where the air-in temperature comes from

The recuperator gives back most of the exhaust heat: air-in = 25 + 0.88 × (stack − 25) at 6000 ln/min, falling to 0.83 at 8000 ln/min and more. When the heaters are on, the air heater lifts the air-in towards its target (300 °C at start, 500 hot standby, 580 fuel switch, 600 part load), no faster than the plant's 60 °C per hour and no more than its 40 kW allow. If the recuperator is already hotter than the target, the 3-way bypass valve holds the target, as on the plant. Air-heater power is then a true energy balance: air flow × heat capacity × temperature lift ÷ 0.89 efficiency.

Try it: add up one minute

Hot standby: stack 505 °C, air-in 500 °C, air 6000 ln/min, heaters on, no gas, no current.

Heat term	°C per hour
Air (5 °C cooler than stack)	-1.6
Hot-box heaters	+7.6
Natural gas	0.0
Current	0.0
Loss to room	-5.8
Sum (°C per hour)	**+0.2**

≈ 0 °C per hour: the stack stays where it is.

Part load: stack 650 °C, air-in 600 °C, air 9000 ln/min, heaters on, gas 137 ln/min, 6.4 A.

Heat term	°C per hour
Air (50 °C cooler, 1.8 × more air)	-29.0
Hot-box heaters	+7.6
Natural gas	+52.9
Current	+40.6
Loss to room	-7.6
Sum (°C per hour)	**+64.5**

+64 °C per hour: the stack keeps warming.

The hot standby sum is why the twin settles near 505 °C by itself: the heat in and out cancel there. The part load sum shows why the twin keeps warming in a long part load. The plant never stayed in part load longer than 44 minutes, so where it would really settle is not in the data.

Two extra physics-shaped pieces

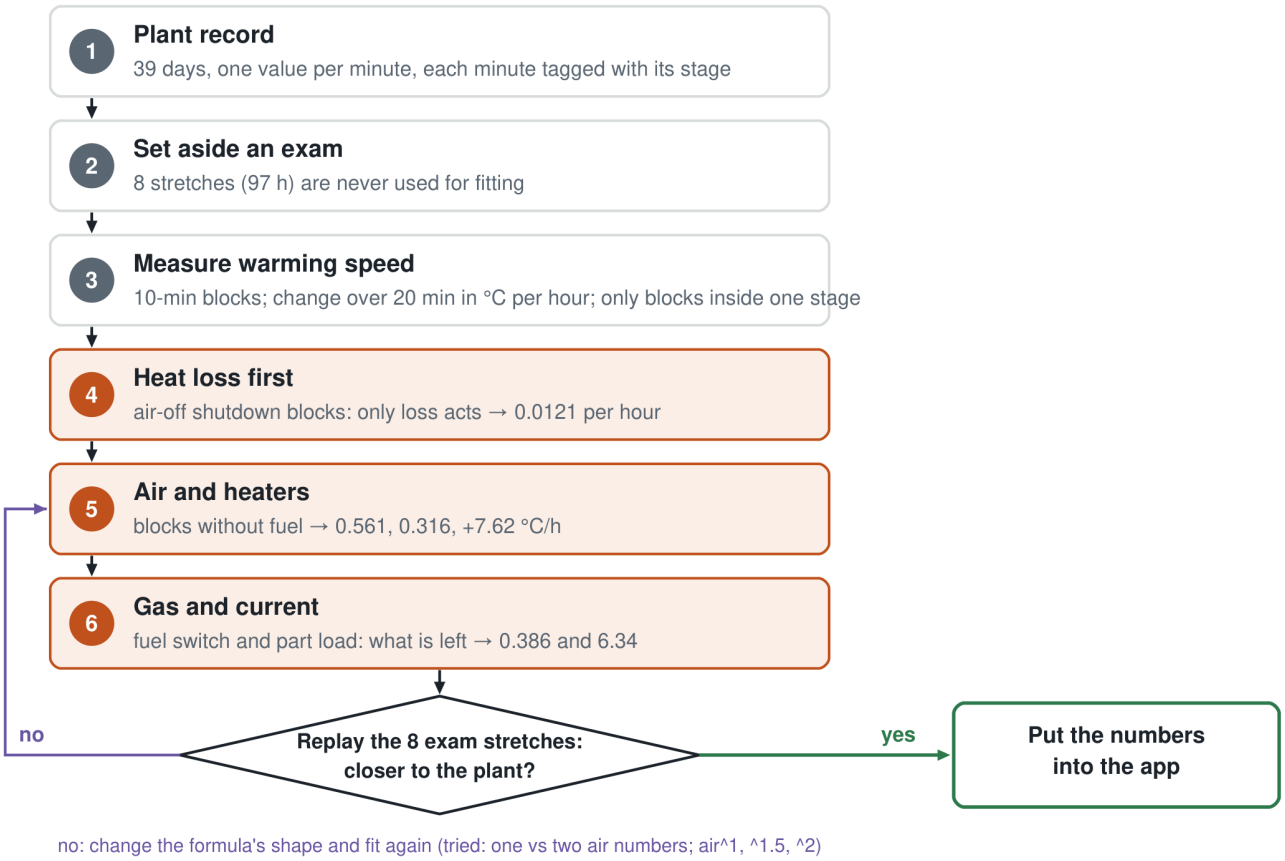
Cooling after hot standby. When the plant leaves hot standby with heaters off, the twin uses a cooling curve fitted to 5 real cool-downs: speed = 0.0605 × (air ÷ 6000)^1.65 × u × (u ÷ 475)^0.5, where u = stack − 25 °C, ramping in over a few minutes. It has the shape of physics but is a fitted curve. **Air-heater inlet.** The heavy recuperators lag the stack by about 32 minutes, and gas adds 0.85 °C per ln/min.

IF YOU REMEMBER ONE THING

Six numbers, all measured from the plant, decide how fast the stack warms or cools.

How the numbers were fitted

Fitting means turning the dials of the formula until its predicted warming speed sits as close as possible to the warming speed the plant really had, minute after minute. Then the result is tested on plant stretches it never saw.



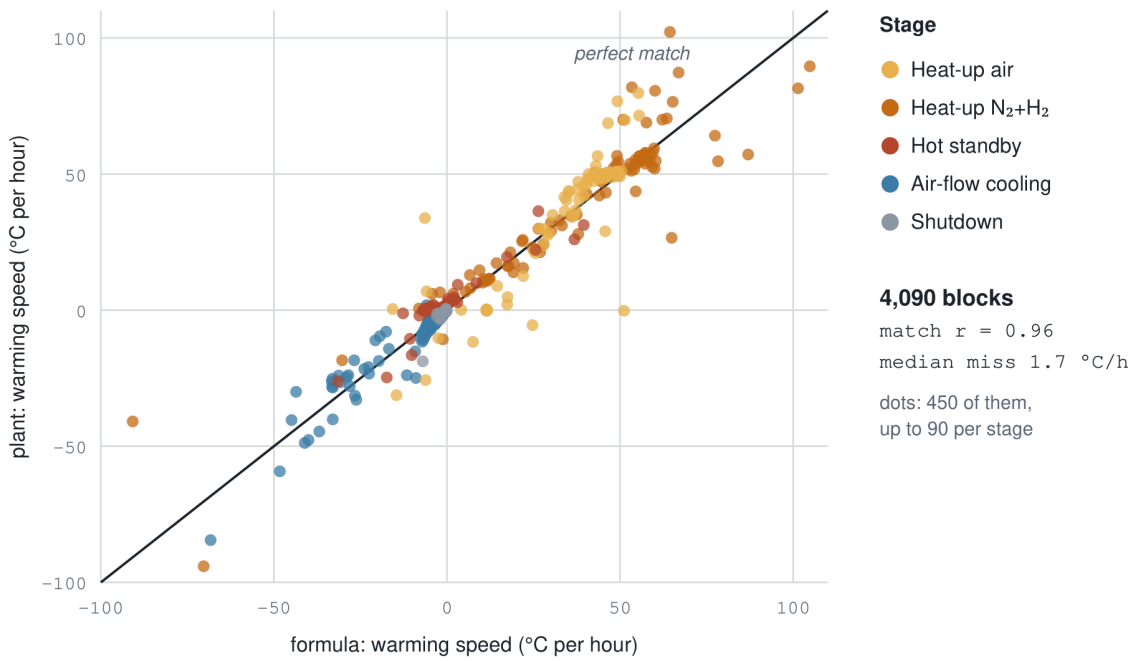
The six steps, then the exam. Steps 4 to 6 each fix only the numbers their kind of data can reveal.

Why in this order

Each step uses data where only a few effects are active, so the numbers cannot hide behind each other. With the air off and heaters off (step 4), only the loss to the room is left, so the loss number is clean. Without fuel (step 5), only air and heaters act. Only then are gas and current (step 6) fitted, on what the first steps cannot explain.

What "closest" means

The method is **least squares**: square every miss between formula and plant, add them up, and pick the numbers that make the sum smallest. Two guards were added. Every stage counts the same, so the long hot standby hours do not drown out the short heat-ups. And no number may go below zero, so a heater can never be fitted as a cooler ("non-negative least squares").



Each dot is one 10-minute block of real plant operation without fuel. Dots near the dashed line mean the formula predicts the plant's warming speed well.

The exam

Eight plant stretches (97 hours: 2 cold starts, 3 standby-to-load runs, 2 air-flow cool-downs, 1 air-off shutdown) were kept out of every fit. The twin is fed the plant's own knob settings minute by minute and its stack temperature is compared with the plant's. Average miss: **56.4 °C before, 17.5 °C after** this fit. When a shape failed the exam it was changed and refitted; that is how the air term became two numbers (warming and cooling) and got the 1.5 power.

Other fitted piece	Data used	Search method	Test result
Recuperator lag and gas gain (air-heater inlet)	All 39 days minus the 8 exam stretches	Simulate, then let a search walk downhill (Nelder–Mead) over 5 numbers	Inlet miss 23.5 → 12.6 °C on held-out minutes
Cooling after hot standby	5 free plant cool-downs (6000–10000 ln/min)	Same search over 4 numbers	Twin replay miss 12.2 → 5.1 °C; each run left out in turn: 10.8 °C
Stage-change glide timing	Every clean stage change on the 39 days	Median curve per sensor, 10 % and 90 % crossing	19.9 → 13.3 % of the move, leaving one change out each time

IF YOU REMEMBER ONE THING

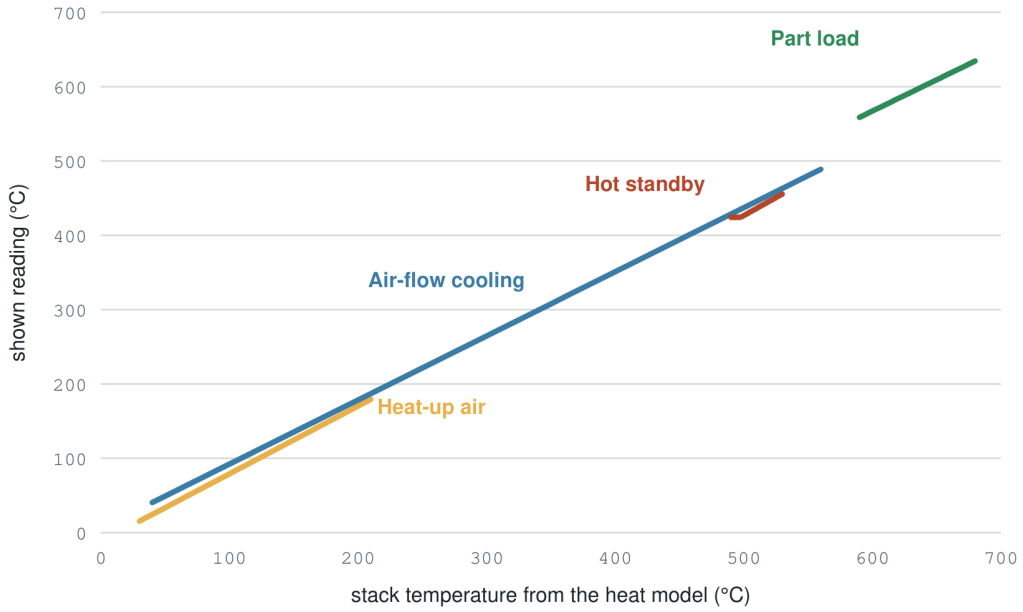
The numbers were chosen so the formula matches 4,090 real 10-minute blocks, then checked on 8 stretches it had never seen.

The parts that come straight from data

For 96 sensors the twin does no physics at all. It asks: "in this stage, at this stack temperature, what did the plant usually read?"

One line per sensor per stage

For each sensor and each of the 8 stages, a script draws the best straight line through the plant minutes: reading = a + b × stack temperature. The line is used only if stack temperature explains at least a quarter of the sensor's ups and downs in that stage ($R^2 \geq 0.25$). Otherwise the twin shows the stage's middle value (median). Either way the value is kept between the lowest 2 % and highest 2 % the plant showed in that stage.



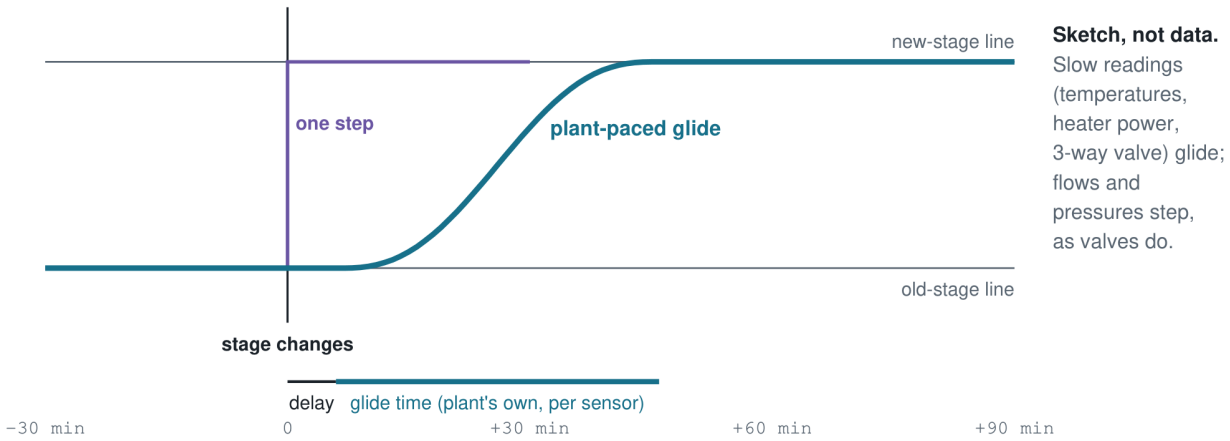
Sensor TIR 2.02: air leaving the air pre-heaters.

Each stage has its own line, fitted on that stage's minutes only, and cut off at the lowest and highest values the plant showed there.

Real lines for one sensor, drawn from the twin's own table. The heat model moves the stack temperature; the line turns it into this sensor's reading.

Kind of reading	Count	Where its value comes from
Stage line (follows stack temperature)	358 of 768 sensor × stage pairs	a + b × stack temperature, plant range limits
Stage median (flat)	410 of 768 pairs	The plant's middle value for that stage
Flows and currents	11 sensors	The knob setting itself
Knob defaults per stage	8 stages	Plant medians: e.g. part load air 9000, gas 137, POx air 312 ln/min, 6.4 A

No jumps at stage changes



Sketch, not data. Slow readings (temperatures, heater power, 3-way valve) glide; flows and pressures step, as valves do.

When the stage changes, the table switches to the new stage's line. Slow readings glide there instead of jumping.

The glide timing is measured too. For 4 kinds of stage change where at least 3 sensors moved clearly on the plant, each of 61 slow sensors has its own delay and glide time from the plant. Other stage changes use the sensor's median glide time. The stack's own warming speed also blends over the plant's bend time.

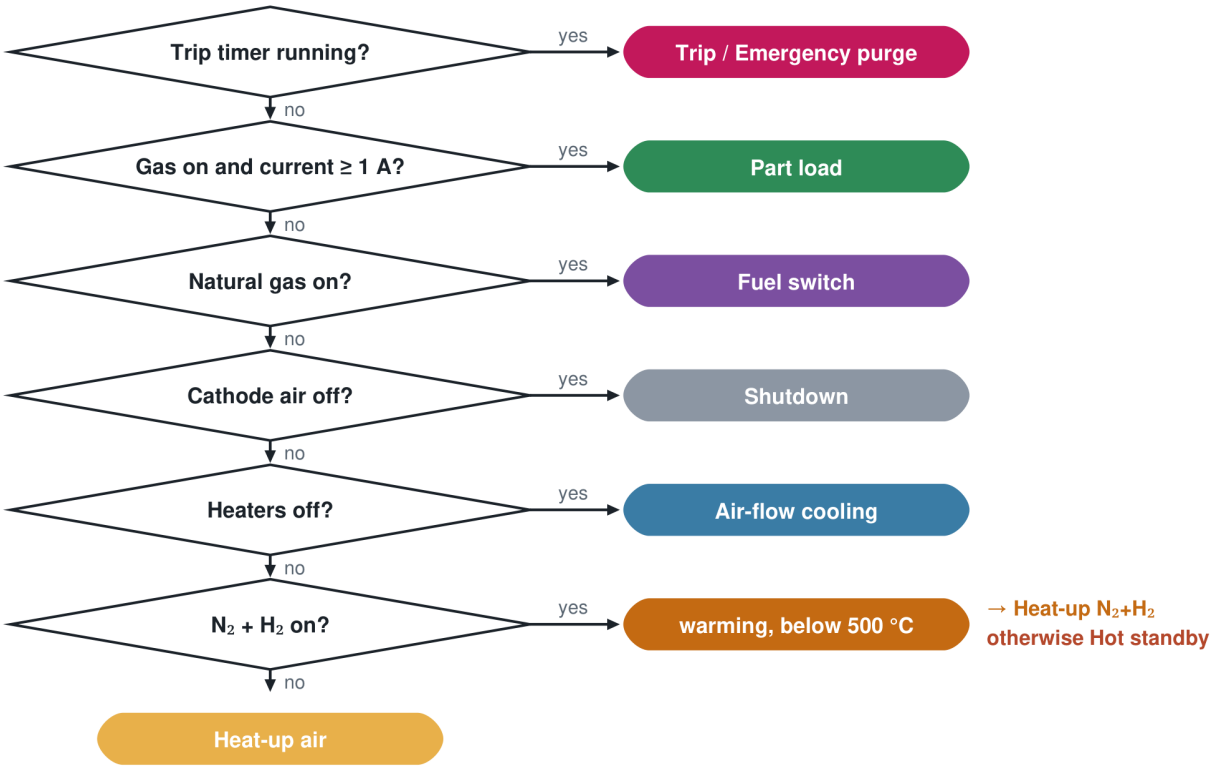
Rebuilt by
`05_twin_2d/tools/prep.py → data/sensors.js and tools/bends.py → data/bends.js`

IF YOU REMEMBER ONE THING

Outside the stack and air heater, the twin repeats what the plant did. It cannot show something the plant never did.

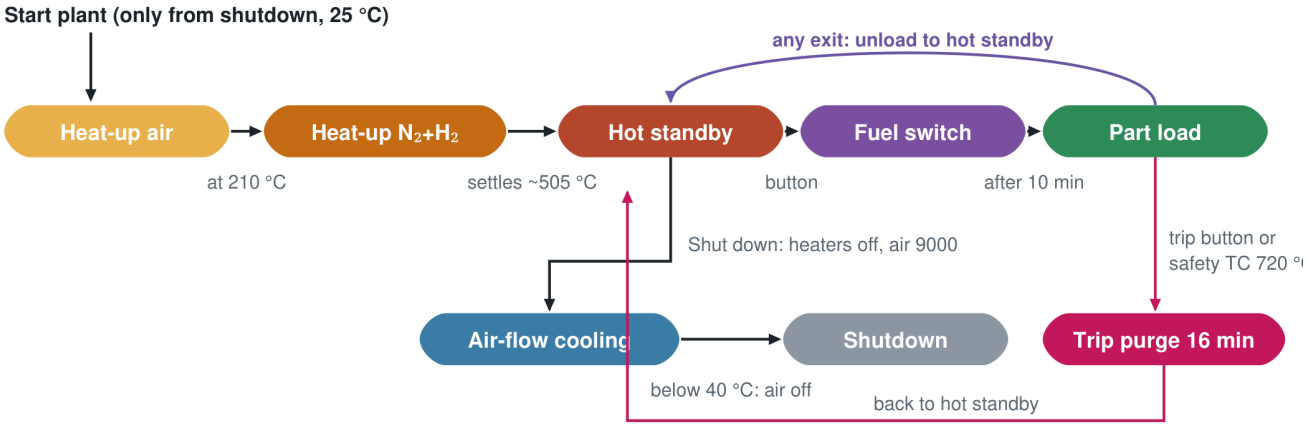
Stages and rules: plant practice, not physics

The twin never guesses the stage from sensors. It reads it straight from the knob settings, by asking six questions in order.



The first "yes" names the stage. Heat-up N_2+H_2 and hot standby share the same knobs; the twin tells them apart by whether the stack is still warming below 500 °C.

Which moves are allowed



Buttons only allow the moves the plant makes. Fuel switch needs hot standby, part load comes 10 minutes after the gas, and part load always ends in hot standby.

Rule	Value	Source
Start plant	Only from shutdown; stack set to 25 °C first	Your decision (v22)
N_2 + H_2 on	Above 210 °C without gas	Plant practice

Rule	Value	Source
Fuel switch → part load	After 10 min	Plant median gas → first current
Leaving part load	Gas stays 10 min at 0 A, then hot standby	Plant, 19 and 30 March
Trip purge	16 min, only from hot standby or part load	Plant purges 15–17 min; your rule
Safety trip	Safety thermocouple TIRZ 8.1.3 at 720 °C	Plant trip limit
Shut down	Air 9000 until the stack is below 40 °C	Plant cool-down after trip 8
Stack keeper (twin only)	Above 660 °C: more air, then lower air-heater target	Not in plant data

IF YOU REMEMBER ONE THING

The stage is read from the knob settings, and the buttons only allow the moves the plant makes.

How close it gets, and where it is weak

Against plant stretches it never saw, the twin's stack temperature is off by **17.5 °C on average**. Every fitted piece was checked the same way: fit on most of the data, test on the rest.



Average miss before and after each fit. Each row has its own scale, shown at its two ends.

Other check	Result
Part load: stack over the 10 plant part-load runs (8–44 min)	4.2 °C average miss when fitted (v12); 10.8 °C since the smooth stage-change blend (v28)
Part load: all 111 readings	Median miss 7.7 % of each sensor's range
Each stage of the 8 exam stretches	Within ±20 °C on average, except air-flow cooling at 9000 ln/min

Known weak spots

Where	What happens	Why
Air-flow cooling at 9000 ln/min	One exam stretch is off by 74 °C	The recuperator was weaker on that day than on the others
Full shut-down cool-down	Twin takes about 34 h, plant took 18 h (664 → 41 °C)	Cooling at high air is too slow in the heat balance
Part load longer than 44 min	Twin keeps warming towards ~759 °C	The plant never ran that long, so there is no data; the stack keeper is a twin-only guess
Air-heater power in part load	Twin too low; plant draws 15–23 kW late in part load	The plant's extra power does not show up as warmer air
The 96 look-up sensors	Show typical plant values, not cause and effect	No physics behind them; a fault or a new setting is not predicted
Cell voltage	Looked up per stage	No electrochemical model (voltage from current and temperature) yet

IF YOU REMEMBER ONE THING

Good enough to follow the plant's stack within about 18 °C on average. Weakest at high-air cooling and long part load, and blind to anything the plant never did.